

Sequences — Question Bank

Engineering Mathematics–I (CSE-1001)

BTech 1st Year, IIIT Bhopal

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Instructions

This bank accompanies the note *Sequences*. Work through a part only after you have read the matching sections of the note.

How the parts are meant to be used:

1. **Part A (MCQ)** — rapid revision. Aim for one minute per question. If you cannot answer in one minute, mark it and return to the note.
2. **Part B (True or False)** — every answer must carry a reason or a counterexample. “False” alone earns nothing.
3. **Part C (Short answer)** — the routine computations that make up the bulk of an examination paper.
4. **Part D (Long answer)** — full proofs and multi-step problems, written out as you would in the examination.
5. **Part E (Critical thinking)** — traps, counterexamples and conceptual questions. These separate a good script from an excellent one.

Answers and full worked solutions are issued as a separate document, *Sequences — Answer Key*.

Part A: Multiple Choice Questions

46 questions $\times$ 1 mark

Choose the one correct option.

Q1. A sequence of real numbers is formally a function

- (a) $\mathbb{R} \rightarrow \mathbb{R}$ (b) $\mathbb{N} \rightarrow \mathbb{R}$ (c) $\mathbb{R} \rightarrow \mathbb{N}$ (d) $\mathbb{N} \rightarrow \mathbb{N}$

Q2. Which statement is correct?

- (a) A sequence is the same thing as the set of its values
(b) Order does not matter in a sequence
(c) The sequence $1, 1, 1, \dots$ has infinitely many terms, but its set of values has one element
(d) Every sequence has finitely many terms

Q3. The general term of $1, 3, 5, 7, \dots$ is

- (a) $2n$ (b) $2n - 1$ (c) $2n + 1$ (d) $n^2 - 1$

Q4. $\lim_{n \rightarrow \infty} \frac{n^2 - 2n}{3n^2 + n}$ equals

- (a) 0 (b) $\frac{1}{3}$ (c) 1 (d) ∞

Q5. The sequence $a_n = 3 + (-1)^n$ is

- (a) convergent to 3 (b) divergent to ∞ (c) oscillating (d) monotonic

Q6. The sequence $a_n = 2^n$ is

- (a) convergent (b) divergent (c) oscillating (d) bounded

Q7. Which sequence is bounded but *not* convergent?

- (a) $\frac{1}{n}$ (b) n (c) $(-1)^n$ (d) $1 - \frac{1}{n}$

Q8. Which sequence is bounded but *not* monotone?

- (a) $\frac{1}{n}$ (b) $\sin n$ (c) n (d) $1 - \frac{1}{n}$

Q9. The sequence $a_n = (-1)^n n$ is

- (a) bounded above only
(b) bounded below only
(c) unbounded above and unbounded below
(d) bounded

Q10. The sequence $a_n = \frac{n}{n+1}$ is

- (a) decreasing and bounded
(b) increasing and bounded
(c) increasing and unbounded
(d) not monotone

Q11. $\lim_{n \rightarrow \infty} a_n = L$ means: for every $\epsilon > 0$ there exists N such that

- (a) $|a_n - L| < \epsilon$ for some $n \geq N$
- (b) $|a_n - L| < \epsilon$ for all $n \geq N$
- (c) $|a_n - L| > \epsilon$ for all $n \geq N$
- (d) $|a_n| < \epsilon$ for all $n \geq N$

Q12. For $a_n = \frac{1}{n}$ and $\epsilon = 0.01$, the smallest N for which $|a_n - 0| < \epsilon$ holds for all $n \geq N$ is

- (a) 10
- (b) 99
- (c) 100
- (d) 101

Q13. Which statement is TRUE?

- (a) Every bounded sequence converges
- (b) Every convergent sequence is bounded
- (c) Every monotone sequence converges
- (d) Every unbounded sequence oscillates

Q14. The Monotone Convergence Theorem guarantees convergence for a sequence which is

- (a) bounded
- (b) monotone
- (c) bounded and monotone
- (d) Cauchy and unbounded

Q15. A monotonically increasing sequence that is not bounded above

- (a) converges
- (b) oscillates
- (c) diverges to $+\infty$
- (d) is Cauchy

Q16. $\lim_{n \rightarrow \infty} n^{1/n}$ equals

- (a) 0
- (b) 1
- (c) e
- (d) ∞

Q17. $\lim_{n \rightarrow \infty} \left(1 + \frac{3}{n}\right)^n$ equals

- (a) e
- (b) e^3
- (c) $3e$
- (d) 1

Q18. $\lim_{n \rightarrow \infty} \frac{n!}{n^n}$ equals

- (a) 0
- (b) 1
- (c) $\frac{1}{e}$
- (d) ∞

Q19. For any $a \in \mathbb{R}$, $\lim_{n \rightarrow \infty} \frac{a^n}{n!}$ equals

- (a) 0
- (b) 1
- (c) a
- (d) ∞

Q20. $\lim_{n \rightarrow \infty} \frac{\log n}{n}$ equals

- (a) 0
- (b) 1
- (c) ∞
- (d) does not exist

Q21. The sequence (r^n) converges precisely when

- (a) $|r| < 1$
- (b) $-1 < r \leq 1$
- (c) $|r| \leq 1$
- (d) all r

- Q22.** $\lim_{n \rightarrow \infty} \frac{\sin n}{n}$ equals
 (a) 1 (b) 0 (c) does not exist (d) ∞
- Q23.** $\lim_{n \rightarrow \infty} \frac{\cos n}{n^2}$ equals
 (a) 0 (b) 1 (c) $\frac{1}{2}$ (d) does not exist
- Q24.** If $a_n > 0$ and $\lim \frac{a_{n+1}}{a_n} = L < 1$, then
 (a) $a_n \rightarrow 1$ (b) $a_n \rightarrow 0$ (c) $a_n \rightarrow L$ (d) (a_n) diverges
- Q25.** For $a_n = \frac{2^n}{n!}$, the ratio $\frac{a_{n+1}}{a_n}$ tends to
 (a) 2 (b) 0 (c) 1 (d) $\frac{1}{e}$
- Q26.** For $a_n = \frac{n!}{n^n}$, $\lim \frac{a_{n+1}}{a_n}$ equals
 (a) 1 (b) e (c) $\frac{1}{e}$ (d) 0
- Q27.** For $a_n = \frac{n}{2^n}$, $\lim(a_n)^{1/n}$ equals
 (a) $\frac{1}{2}$ (b) 1 (c) 2 (d) 0
- Q28.** If the ratio test for sequences gives $L = 1$, the test
 (a) proves convergence
 (b) proves divergence
 (c) is inconclusive
 (d) cannot be applied at all
- Q29.** A subsequence of (a_n) is obtained by
 (a) reordering the terms
 (b) deleting some terms and keeping the rest in order
 (c) inserting new terms
 (d) taking absolute values of the terms
- Q30.** If $a_n \rightarrow L$, then every subsequence of (a_n)
 (a) diverges
 (b) converges to L
 (c) converges to some other limit
 (d) is bounded but need not converge
- Q31.** The sequence $a_n = (-1)^n$ has subsequential limits
 (a) only 1 (b) only -1 (c) 1 and -1 (d) none

Q32. The Bolzano–Weierstrass theorem states that every bounded sequence has

- (a) a limit
- (b) a convergent subsequence
- (c) a divergent subsequence only
- (d) finitely many distinct terms

Q33. (a_n) is a Cauchy sequence if for every $\epsilon > 0$ there is N with

- (a) $|a_n - L| < \epsilon$ for all $n \geq N$
- (b) $|a_m - a_n| < \epsilon$ for all $m, n \geq N$
- (c) $|a_{n+1} - a_n| < \epsilon$ for all $n \geq N$
- (d) $|a_n| < \epsilon$ for all $n \geq N$

Q34. In $\mathbb{R}$, a sequence converges if and only if it is

- (a) bounded
- (b) monotone
- (c) Cauchy
- (d) positive

Q35. Which of the following is NOT a Cauchy sequence?

- (a) $\frac{1}{n}$
- (b) $(c, c, c, \dots)$
- (c) $\frac{n+1}{n}$
- (d) $1 + \frac{1}{2} + \dots + \frac{1}{n}$

Q36. The statement “every Cauchy sequence converges” fails in

- (a) $\mathbb{R}$
- (b) $\mathbb{Q}$
- (c) both
- (d) neither

Q37. Which of the following tends to 0?

- (a) $\frac{n!}{2^n}$
- (b) $\frac{2^n}{n!}$
- (c) $\frac{n^n}{n!}$
- (d) $\frac{n}{\ln n}$

Q38. The limit of a convergent sequence is

- (a) not unique
- (b) unique
- (c) always 0
- (d) always a term of the sequence

Q39. For $a_n = 1 - \frac{1}{n}$, the limit 1 is

- (a) a term of the sequence
- (b) the least upper bound, never attained
- (c) a lower bound of the sequence
- (d) not a bound at all

Q40. For $a_1 = 100$ and $a_{n+1} = \frac{1}{2}a_n + 100$, the limit is

- (a) 100
- (b) 150
- (c) 200
- (d) ∞

Q41. $\lim_{n \rightarrow \infty} \frac{3n-1}{1+2n}$ equals

- (a) $\frac{3}{2}$
- (b) 3
- (c) $\frac{1}{2}$
- (d) 0

- Q42.** The sequence $a_n = \left(\frac{n}{n-1}\right)^2$, $n \geq 2$, converges to
- (a) 0 (b) 1 (c) e (d) ∞
- Q43.** The sequence $a_n = 1 + \frac{(-1)^n}{n}$ is
- (a) monotonically increasing
 (b) not monotone, but convergent to 1
 (c) divergent
 (d) unbounded
- Q44.** The sequence $a_n = \sin n$ is
- (a) convergent to 0
 (b) bounded and divergent
 (c) unbounded
 (d) monotone
- Q45.** The sequence $a_n = (-1)^{n-1}$ is
- (a) convergent to 1
 (b) convergent to -1
 (c) oscillating and bounded
 (d) unbounded
- Q46.** If (a_n) is unbounded, then (a_n)
- (a) must diverge
 (b) may still converge
 (c) must oscillate
 (d) must be monotone

Part B: True or False — with justification	10 questions $\times$ 1 mark
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State whether each is true or false. **A verdict without a reason or a counterexample earns no marks.**

- Q1.** Every bounded sequence is convergent.
- Q2.** Every convergent sequence is bounded.
- Q3.** If $a_n \rightarrow 0$, then (a_n) is monotone.
- Q4.** A monotone sequence is always bounded.
- Q5.** If $(|a_n|)$ converges, then (a_n) converges.
- Q6.** If (a_n) has a convergent subsequence, then (a_n) converges.
- Q7.** If $|a_{n+1} - a_n| \rightarrow 0$, then (a_n) is a Cauchy sequence.

Q8. A sequence can have two different limits.

Q9. If $a_n > 0$ for every n and $a_n \rightarrow L$, then $L > 0$.

Q10. Every Cauchy sequence of rational numbers converges to a rational number.

Part C: Short Answer Questions

20 questions $\times$ 3 marks

Q1. Write the general term of each sequence:

(i) $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$

(ii) $2, 5, 8, 11, 14, \dots$

(iii) $3, 6, 12, 24, 48, \dots$

Q2. Examine the convergence of $a_n = \frac{n^2 - 2n}{3n^2 + n}$.

Q3. Examine the convergence of $a_n = \frac{3n - 1}{1 + 2n}$.

Q4. Examine the convergence of $a_n = 3 + (-1)^n$, and name the behaviour.

Q5. Is the sequence $1, 3, 5, 7, \dots, (2n - 1), \dots$ bounded? Justify your answer fully.

Q6. Show that $a_n = \frac{n}{n + 1}$ is monotonically increasing and bounded. What does the Monotone Convergence Theorem then give?

Q7. Show that $a_n = \frac{2^n}{n!}$ is monotonically decreasing.

Q8. Using the ϵ - N definition, prove that $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$.

Q9. Using the ϵ - N definition, prove that $\lim_{n \rightarrow \infty} \frac{2n}{n + 3} = 2$.

Q10. For $a_n = \frac{1}{n}$, find the least N such that $|a_n - 0| < 0.001$ for all $n \geq N$.

Q11. Evaluate $\lim_{n \rightarrow \infty} \left(1 + \frac{a}{n}\right)^n$ and hence $\lim_{n \rightarrow \infty} \left(1 + \frac{3}{n}\right)^n$.

Q12. Evaluate $\lim_{n \rightarrow \infty} n^{1/n}$ and $\lim_{n \rightarrow \infty} 5^{1/n}$.

Q13. Use the Sandwich Theorem to evaluate $\lim_{n \rightarrow \infty} \frac{\sin n}{n}$.

Q14. Use the Sandwich Theorem to show that $\lim_{n \rightarrow \infty} \frac{(-1)^n}{\sqrt{n}} = 0$.

Q15. Exhibit two subsequences of $a_n = (-1)^n$ with different limits, and state the conclusion.

- Q16.** Show, directly from the definition, that $a_n = n$ is not a Cauchy sequence.
- Q17.** Show, directly from the definition, that $a_n = \frac{1}{n}$ is a Cauchy sequence.
- Q18.** Apply the ratio test to $a_n = \frac{3^n}{n!}$.
- Q19.** Apply the root test to $a_n = \frac{1}{n^n}$ and to $a_n = \left(\frac{3n+1}{4n+2}\right)^n$.
- Q20.** State the Monotone Convergence Theorem, and illustrate it with one example of your own.

Part D: Long Answer Questions

12 questions $\times$ 6 marks

- Q1.** Define the limit of a sequence using the ϵ - N definition. Prove from the definition that $\lim_{n \rightarrow \infty} \frac{n+1}{n} = 1$, and find an explicit N that works for $\epsilon = 0.01$.
- Q2.** Prove that every convergent sequence is bounded. Is the converse true? Justify with a counterexample, and state the form of this theorem that is actually used to prove divergence.
- Q3.** Prove that a convergent sequence has a unique limit.
- Q4.** A patient takes 100 mg of a drug each morning, and the body eliminates half of whatever is present over 24 hours. With a_n the amount just after the n -th dose,
- $$a_1 = 100, \quad a_{n+1} = \frac{1}{2}a_n + 100.$$
- Prove that (a_n) is increasing and bounded above by 200, deduce that it converges, and find the limit. Interpret the answer for the patient.
- Q5.** Let $a_1 = 2$ and $a_{n+1} = \frac{1}{2} \left(a_n + \frac{2}{a_n} \right)$.
- (i) Show that $a_n \geq \sqrt{2}$ for all $n \geq 2$.
 - (ii) Show that (a_n) is decreasing.
 - (iii) Deduce convergence and find the limit.
 - (iv) Compute a_2, a_3, a_4 correct to six decimal places and comment on the speed of convergence.
- Q6.** Let $a_1 = 1$ and $a_{n+1} = \sqrt{2 + a_n}$. Show that (a_n) is increasing and bounded above by 2, and find its limit.
- Q7.** Examine the convergence of each of the following sequences, classifying each as convergent, divergent or oscillating:

$$\begin{array}{llll} \text{(i)} \ (-1)^{n-1} & \text{(ii)} \ 1 + \frac{2}{n} & \text{(iii)} \ \frac{3n-1}{1+2n} & \text{(iv)} \ 2n \\ \text{(v)} \ 1 + \frac{(-1)^n}{n} & \text{(vi)} \ \left(\frac{n}{n-1} \right)^2 & \text{(vii)} \ \sin n & \end{array}$$

Q8. State the Sandwich Theorem. Use it to evaluate

$$\begin{aligned} & \text{(i)} \quad \lim_{n \rightarrow \infty} \frac{\cos n}{n^2} \qquad \text{(ii)} \quad \lim_{n \rightarrow \infty} \frac{n + \sin n}{n} \\ & \text{(iii)} \quad \lim_{n \rightarrow \infty} \left(\frac{1}{n^2 + 1} + \frac{1}{n^2 + 2} + \cdots + \frac{1}{n^2 + n} \right) \end{aligned}$$

Q9. Explain the three types of divergence of a sequence, giving two examples of each, and indicate roughly how the terms are placed on an index plot in each case.

Q10. Define a Cauchy sequence. Prove that every convergent sequence is Cauchy. State the converse for $\mathbb{R}$, and explain with an explicit example why the converse fails in $\mathbb{Q}$.

Q11. State the ratio test and the root test for sequences. Apply the appropriate test to each of

$$\text{(i)} \quad \frac{2^n}{n!} \qquad \text{(ii)} \quad \frac{n!}{n^n} \qquad \text{(iii)} \quad \frac{n}{2^n} \qquad \text{(iv)} \quad \left(\frac{3n+1}{4n+2} \right)^n$$

and in each case state the conclusion.

Q12. Write down the growth hierarchy $\ln n \prec n^p \prec a^n \prec n! \prec n^n$ ($p > 0$, $a > 1$), explain what the symbol $\prec$ means, and use it to evaluate

$$\lim_{n \rightarrow \infty} \frac{\ln n}{n}, \quad \lim_{n \rightarrow \infty} \frac{n^{100}}{2^n}, \quad \lim_{n \rightarrow \infty} \frac{2^n}{n!}, \quad \lim_{n \rightarrow \infty} \frac{n!}{n^n}, \quad \lim_{n \rightarrow \infty} \frac{n!}{2^n}.$$

Part E: Critical Thinking Questions

8 questions $\times$ 5 marks

These reward precision. Read each statement exactly as written.

Q1. A student defines: “a sequence (a_n) is bounded if there exists a real number M such that $a_n < M$ for all n .” Show by an explicit example that this definition is incomplete, state what property it actually captures, and write the correct definition of a bounded sequence.

Q2. Both $a_n = \frac{1}{n}$ and $a_n = n$ give $L = 1$ in the ratio test. One converges and the other diverges. Explain precisely why the test is inconclusive at $L = 1$, and determine the behaviour of each sequence by another method.

Q3. Give, with justification, an example of a sequence which is

- (i) bounded, not monotone, and convergent;
- (ii) bounded, not monotone, and divergent;
- (iii) monotone, unbounded;
- (iv) neither bounded nor monotone.

Q4. For the recursion $a_1 = 1$, $a_{n+1} = 2a_n$, a student argues: “Let $L = \lim a_n$. Taking limits, $L = 2L$, hence $L = 0$.” The true behaviour is $a_n = 2^{n-1} \rightarrow \infty$. Identify exactly which step of the argument is invalid, and state the principle that the student has violated.

- Q5.** Construct a sequence whose set of subsequential limits is exactly $\{0, 1, 2\}$, and prove that your sequence works. Is such a sequence bounded? Does it converge?
- Q6.** Prove that if $a_n \geq 0$ for all n and $a_n \rightarrow L$, then $L \geq 0$. Show by an example that the strict inequality $a_n > 0$ for all n does *not* force $L > 0$, and explain why strictness is lost in the limit.
- Q7.** Exhibit a sequence of rational numbers which is Cauchy but has no limit in $\mathbb{Q}$. Explain what property of $\mathbb{R}$ this shows is missing from $\mathbb{Q}$, and why that property matters for calculus.
- Q8.** Suppose the subsequence of even-indexed terms (a_{2n}) and the subsequence of odd-indexed terms (a_{2n-1}) both converge to the same limit L . Prove that $a_n \rightarrow L$. Show by an example that the conclusion fails if the two limits are different.